12TH CHEMISTRY · CHAPTER 22 · CHAPTER TEST

Hydroxy Compounds

Chapter-wise test: 6 MCQs, 5 of 6 short questions, 1 long question

Class: 12th (HSSC Part II)	**Subject:** Chemistry	**Total Marks:** 26
Chapter: 22	**Time Allowed:** 45 minutes	**Name:** _____

OBJECTIVE PART (6 × 1 = 6)

Q.1 Choose the correct answer.

1. Which is a tertiary alcohol?

(a) Ethanol (b) Propan-2-ol
(c) Propan-1-ol (d) 2-Methylpropan-2-ol

2. The strongest acid among these is:

(a) Ethanol (b) Phenol
(c) Methanol (d) Water

3. Oxidation of a secondary alcohol gives:

(a) Ketone (b) Aldehyde
(c) Carboxylic acid (d) Alkene

4. Acidified potassium dichromate changes from orange to:

(a) Colourless (b) Purple
(c) Green (d) Yellow

5. Phenol with bromine water gives a precipitate which is:

(a) Yellow (b) White
(c) Black (d) Pink

6. Dehydration of an alcohol gives an:

(a) Alkane (b) Ester
(c) Alkene (d) Aldehyde

SUBJECTIVE PART — SHORT QUESTIONS (5 × 2 = 10)

Q.2 Write short answers to any FIVE (5) questions.

i. Differentiate between monohydric and polyhydric alcohols.
ii. Why is ethanol a weaker acid than water?
iii. How is the reaction with sodium used as a test for the hydroxy group?
iv. Why are tertiary alcohols not detected by acidified dichromate?
v. Why is phenol nitrated more easily than benzene?
vi. What happens when phenol reacts with sodium hydroxide?

LONG QUESTION (2 × 5 = 10)

Q.3 Answer both parts.

(a) Describe six methods for the synthesis of alcohols.
(b) How is phenol prepared from phenylamine? Explain the acidity of phenol.

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